Leticia Diniz

linkedin.com/in/letiicia-dnz · leticiadiniz2028@u.northwestern.edu · leticiadiniz.site | +1(733) 858-2243

Education

Northwestern University, Evanston, IL *Anticipated June 2028*
B.S. Computer Engineering | Segal Design Certificate
Relevant Coursework: Smart Electronics Design, Microcontroller System Design, Interdisciplinary Product Design I

Skills

Mechanical Design & Manufacturing: CAD & DFM (Onshape, SolidWorks / Fusion 360), tolerance and fit for impact loading, mount and fastener design, 3D printing, subtractive machining, rapid prototyping
Robotics & Mechatronics: Hardware system architecture, actuator and motor selection, motor drivers & ESCs, drivetrain and weapon-system design, closed-loop and PID control, sensor interfacing, ROS 2
Electronics & Software: PCB design (Eagle), analog signal conditioning, low-voltage power electronics, I^2C / SPI / UART, soldering & rework, oscilloscope debugging, Embedded C/C++, Python, MATLAB (Simulink), Git, Linux
Languages: Portuguese (native), English (fluent), Spanish (basic), Hebrew (basic)

Professional Experience

Robotic Matter Lab, Northwestern University *Oct 2025 – Present*
Undergraduate Researcher

- Designing and building a custom automated doctor-blade coating system for scalable fabrication of thin-film soft actuators, replacing a manual process with a precision motion stage, controlled thermal zone, and UV-curing stage in one instrument
- Specified the mechanical process envelope — 30–90° blade angle, 25–100 μm gap control, 5–100 mm/s deposition speed — and designed the fixturing and adjustment mechanisms that hold those tolerances repeatably

Northwestern University Robotics Club (NURC) *Jan 2025 – Present*
Combat Robotics Co-Leader

- Co-lead a 8 -person sub team and a budget building competition robots in the 3 lb and 30 lb weight classes
- Designed a scratch-built 3 lb combat robot end to end in CAD — chassis, drivetrain, and a steel spinning weapon reaching 100 MPH tip speeds;
- Upgraded a 3 lb kit robot with a custom saw weapon, redesigning blade geometry and impact-bearing mounts in CAD and manufacturing them with additive and subtractive processes to improve reliability under repeated impacts
- Building the full electronics and drive control system for a 30 lb combat robot: motor driver and ESC selection, power distribution, and custom radio-control channel mapping

Deco.cx, Rio de Janeiro, Brazil *Apr 2024 – Sep 2024*
Software Development Intern

- Built and optimized internal developer tooling for AI-assisted web development workflows, cutting build and deployment time ~40% and improving release reliability across multiple projects
- Shipped customizable backend-driven web platforms for Brazilian retail clients (Stone, BAW) with cross-functional teams, reducing client-specific deployment overhead

Patents

Accessible pH Meter — Federal Institute of Paraíba *Apr 2020 – Dec 2022*

- Invented and patented a low-cost Arduino-based pH meter that converts acidity readings into real-time audio feedback, enabling independent laboratory use by visually impaired chemists
- Designed and built the complete physical device: a 3D-printed enclosure making it fully self-contained and portable, around a custom PCB with analog conditioning for a BNC pH electrode and embedded firmware

SARA — Federal Institute of Paraíba *Apr 2021 – Dec 2021*

- Co-developed and patented a backend-driven automation platform for repetitive academic administrative workflows, cutting task completion time ~5× (≈80%) through system integration and process redesign

Projects

PoweReg — Automated Oxygen Regulator, Northwestern University *Sep 2024 – Dec 2024*
Team Lead

- Led a 4-person team building an automated oxygen-tank setup device for Lurie Children's Hospital, replacing a manual process users described as taking over a minute with 34 s of hands-free actuation at an $80.67 bill of materials
- Ran the material and DFM iteration through destructive bench testing: raised 3D-print infill from 15% to 50% after the printed valve cylinder chipped under motor load, and replaced a PLA pin holder that deflected under torque with aluminum

ALS Vocal Communicator, Federal Institute of Paraíba *Apr 2021 – Dec 2021*

- Built an Arduino-based vocal communicator for patients with ALS, integrating sensor-based interaction and real-time signal processing to restore speech capability; presented to 700+ attendees